Are LSTM Uncertainty Estimates Calibrated?
A Post-Hoc Diagnostic Study of Probabilistic WTI Crude Oil Forecasts

Peter Prendergast

ABSTRACT

Our earlier analysis suggested that the LSTM's learned σ_t was "calibrated and regime-aware," but that claim was not tested formally. This report studies the standardised residuals of two LSTM variants and a GARCH(1,1) benchmark using goodness-of-fit, dependence, and coverage diagnostics. The results suggest that the pure NLL LSTM is only partly calibrated, the composite-loss model distorts σ_t, and a Student-t residual law improves fit for all three models.

CONTENTS

I. INTRODUCTION

Forecasting next-day returns in commodity markets is difficult because the signal-to-noise ratio is extremely low. When models are trained under standard loss functions such as MSE or Gaussian negative log likelihood (NLL), the optimal prediction is typically close to zero. In our group work, this was referred to as the "flat-line problem." It is not necessarily a failure of modelling, but a consequence of the structure of the signal. The conditional mean of daily log returns is very close to zero, so aggressive departures are penalised heavily.

To address this problem, our project introduced a composite three-component loss function that combines Gaussian NLL with a directional penalty and a variance-collapse penalty. Under this composite loss, the LSTM produced non-trivial directional predictions (52.3% accuracy versus 47.1% under pure NLL) and a tenfold increase in prediction spread. We also interpreted the learned σ_t as "calibrated and regime-aware," based on two pieces of informal evidence: the unconditional average of σ_t (≈ 0.025) is close to the empirical return standard deviation (≈ 0.023), and the uncertainty bands visibly widen during volatile periods.

However, we did not complete a formal calibration test. Matching unconditional moments is a necessary but not sufficient condition for probabilistic calibration. The composite loss is also not a proper scoring rule in the sense of Gneiting and Raftery [9]. The Gaussian NLL is proper because it is the logarithmic score for the Gaussian family. But the composite loss adds terms $\mathcal{L}_{\text{dir}}$ and $\mathcal{L}_{\text{var}}$ that depend on μ_t through non-probabilistic objectives. Their expectation under the true distribution is not minimised at $F = G$, so the model is no longer trained purely to report the true predictive distribution. As a result, σ_t may absorb part of that distortion.

Calibration of neural network uncertainty estimates is a known challenge in the broader deep learning literature [11], and the question of whether heteroscedastic regression networks produce calibrated variance estimates has received less attention in the financial forecasting context. It is also natural to consider using σ_t for position sizing in a simulated trading strategy. Before σ_t can be trusted for use in risk management, we must establish whether it is calibrated.

This report addresses that gap through a focused calibration study of the models developed in the earlier analysis. The central research question is whether the uncertainty estimates produced by our LSTM models are statistically calibrated and how they compare with a classical GARCH benchmark.

The analysis is organised around three hypotheses:

1. the pure NLL LSTM should be better calibrated than the composite decision-aware model because NLL is a proper scoring rule;

2. both models are likely to exhibit heavy-tailed residuals relative to the Gaussian assumption, so a Student-t predictive law should improve fit;

3. the LSTM's volatility estimates may not outperform a GARCH(1,1) benchmark once evaluated using formal density-forecast diagnostics.

II. DATA ANALYSIS

A. Dataset and target variable

The analysis uses the dataset prepared for the earlier study: daily observations for the WTI crude oil futures contract from January 2017 to March 2026. The target variable is the one-day-ahead log return,

$$r_t = \log P_t - \log P_{t-1}, \tag{1}$$

where P_t denotes the daily settlement price. The 51 predictors used by the LSTM comprise lagged returns, rolling volatility measures, volume/open-interest features, market microstructure variables, and calendar indicators. Because the predictive mean of r_t is economically small, the main object of interest for this report is not μ_t itself but the accompanying volatility estimate σ_t.

B. Train/validation/test split

To preserve temporal ordering and avoid look-ahead bias, I use a walk-forward split: the training set runs from 31 Jan 2017 to 30 Dec 2022 (1,235 observations), the validation set from 3 Jan 2023 to 28 Jun 2024 (312 observations), and the test set from 1 Jul 2024 to 3 Mar 2026 (348 observations).

C. Test-set return distribution

Before assessing model calibration, it is useful to characterise the return distribution itself. The test period exhibits the standard stylised facts of financial returns: near-zero mean, volatility clustering, slight skewness, and excess kurtosis [5]. Table I summarises the basic distributional moments. The high kurtosis implies heavier tails than a Gaussian benchmark would predict, suggesting that a normal predictive density may be misspecified even for a well-calibrated conditional volatility model.

TABLE I. Summary statistics for test-set returns ($n = 348$).

Mean	**Std. dev.**	**Skewness**	**Excess kurtosis**
0.0001	0.0232	−0.28	4.86

D. Models

The report evaluates the two LSTM variants developed in the group work:

1. **Pure-NLL LSTM.** This model outputs (μ_t, σ_t) and is trained by minimising the Gaussian negative log-likelihood,
$$\mathcal{L}_{\mathrm{NLL}} = \frac{1}{2}\log(2\pi\sigma_t^2) + \frac{(r_t - \mu_t)^2}{2\sigma_t^2}, \tag{2}$$
which is a proper scoring rule and therefore incentivises honest distributional forecasts.

2. **Composite-loss LSTM.** This model augments NLL with directional and variance-spread penalties,
$$\mathcal{L}_{\mathrm{comp}} = \mathcal{L}_{\mathrm{NLL}} + \lambda_1\mathcal{L}_{\mathrm{dir}} + \lambda_2\mathcal{L}_{\mathrm{var}}, \tag{3}$$
where $\mathcal{L}_{\mathrm{dir}}$ rewards correct directional calls and $\mathcal{L}_{\mathrm{var}}$ discourages collapse to an unconditional variance forecast. This objective aims to improve trading usefulness, but it sacrifices the strict probabilistic interpretation of σ_t.

In addition, a GARCH(1,1) model is fitted to the training returns as a classical volatility benchmark. GARCH is the standard tool for modelling conditional heteroscedasticity in financial returns and provides a natural reference point [2, 8]. The earlier modelling setup also connects the LSTM's heteroscedastic output to GARCH-style volatility modelling, which makes this comparison especially relevant.

Figure 1 presents the test-set return distribution alongside fitted Gaussian and Student-t density overlays, a Q-Q plot, a P-P plot, and the rolling 60-day realised volatility. The Q-Q plot reveals clear tail departures from the Gaussian model, particularly in the left tail. The rolling volatility panel shows distinct volatility regimes during the test period, providing the visual context for assessing whether the models' σ_t tracks these regimes.

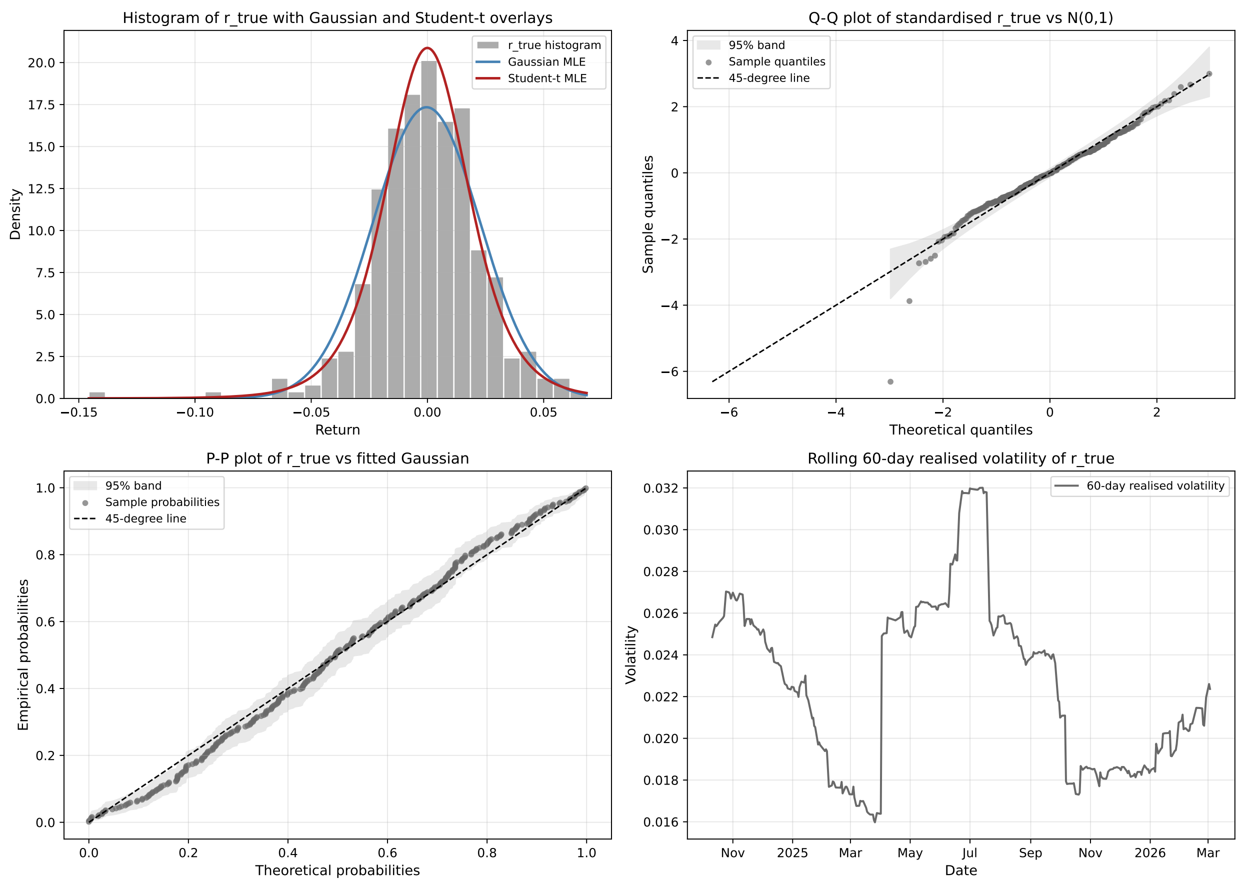

FIG. 1. Test-set return distribution. Top left: histogram with Gaussian and Student-t MLE overlays. Top right: Q-Q plot of standardised r_t against $\mathcal{N}(0,1)$. Bottom left: P-P plot of r_t against the fitted Gaussian. Bottom right: rolling 60-day realised volatility.

III. METHODOLOGY

A. Probabilistic calibration

A probabilistic forecaster that outputs (μ_t, σ_t) is calibrated if for every nominal coverage level $\alpha \in (0,1)$, the empirical fraction of observations falling within the α-prediction interval equals α:

$$\frac{1}{n}\sum_{t=1}^{n} \mathbf{1}[r_t \in (\mu_t - q_\alpha\, \sigma_t,\ \mu_t + q_\alpha\, \sigma_t)] \approx \alpha, \tag{4}$$

where q_α is the $(1+\alpha)/2$ quantile of the assumed predictive distribution.

An equivalent characterisation is provided by the probability integral transform (PIT). If the predictive distribution $F_t(r) = \Phi((r-\mu_t)/\sigma_t)$ is correctly specified, the PIT values

$$u_t = \Phi\left(\frac{r_t - \mu_t}{\sigma_t}\right) \tag{5}$$

should be i.i.d. Uniform$(0,1)$ [7]. Thus, calibration can be assessed through four diagnostics: **distributional fit** (mean-zero, unit-variance standardised residuals), **independence** (no serial dependence in z_t or $|z_t|$), **coverage** (nominal prediction intervals attain their target frequencies), and **PIT uniformity** (PIT values are close to uniform, without U-shapes or inverse-U patterns).

No single test is decisive in isolation. For that reason, this report uses a suite of diagnostics rather than a single summary metric.

B. Statistical tests

The core analysis is based on the standardised residuals:

$$z_t = \frac{r_t - \mu_t}{\sigma_t}. \tag{6}$$

If the Gaussian predictive distribution is well calibrated, then ideally $z_t \overset{i.i.d.}{\sim} \mathcal{N}(0,1)$. Four classes of tests are applied.

a. Moment and shape diagnostics. Sample mean, standard deviation, skewness, and excess kurtosis provide a first-pass assessment. For a calibrated Gaussian density, these should be close to $(0,1,0,0)$. Large deviations immediately indicate bias, under/over dispersion or misspecified tails.

b. Independence diagnostics. Autocorrelation functions of z_t and $|z_t|$ are examined visually, and the Ljung–Box test is applied at lag 10 [12]. Serial correlation in z_t implies mean misspecification; serial correlation in $|z_t|$ implies unmodelled volatility persistence. Either failure indicates that the predictive density has not captured the time-series structure of returns.

c. Distributional goodness-of-fit. The KS test compares the empirical CDF of z_t with the standard normal CDF, while the Anderson–Darling (AD) test emphasises tail discrepancies [1]. These tests complement moment diagnostics by formalising whether the entire shape of the predictive law is plausible.

d. Interval and PIT diagnostics. For nominal levels from 50% to 99%, the empirical coverage of Gaussian prediction intervals is compared with the target coverage. The PIT histogram is also examined: a U-shape indicates under-dispersion, an inverse-U indicates over-dispersion, and skewness indicates location bias [7].

C. GARCH(1,1) benchmark

To benchmark the LSTM's uncertainty estimates against a classical volatility model, the report fits a Gaussian GARCH(1,1) to the training returns,

$$r_t = \varepsilon_t, \qquad \varepsilon_t \mid \mathcal{F}_{t-1} \sim \mathcal{N}(0, h_t), \tag{7}$$

$$h_t = \omega + \alpha\varepsilon_{t-1}^2 + \beta h_{t-1}, \tag{8}$$

where h_t is the conditional variance. The benchmark is intentionally simple: it uses no exogenous features and assumes a zero conditional mean, but it is designed specifically to model volatility clustering. If the deep model's σ_t cannot outperform GARCH on calibration grounds, its additional complexity adds little value for probabilistic forecasting.

Out-of-sample forecasts for the test period are produced by recursively applying Equation (8) and using the validation period as a bridge to preserve the volatility state.

D. Conditional calibration

Unconditional calibration tests may mask regime-dependent miscalibration. If a model is overconfident during volatile periods and overdispersed during calm periods, the aggregate statistics may average to acceptable values. To detect such behaviour, the test set is split into high-volatility and low-volatility subsets using the median of the rolling 60-day realised volatility as the threshold. This method was motivated by Kuleshov et al. [11]. The calibration tests are then applied separately to each subset.

E. Student-t extension

After documenting Gaussian miscalibration, I relaxed the distributional assumption. This follows Bollerslev [3], who introduced the Student-t GARCH specification for financial returns. For each model's standardised residuals z_t, the degrees-of-freedom parameter $\hat{\nu}$ is estimated by maximum likelihood, maximising

$$\ell(\nu) = \sum_{t=1}^{n} \log f_\nu(z_t), \tag{9}$$

where f_ν is the density of a standardised Student-t variable with variance one. The fit is restricted to $\nu > 2$ so that the variance exists. If the heavy tails of returns are the main source of Gaussian miscalibration, then replacing $\mathcal{N}(0,1)$ with $t_\nu(0,1)$ should improve the Q-Q alignment and reduce the rejection rate of the goodness-of-fit tests.

F. Limitations

Several limitations should be noted. Because only 348 test observations are available for evaluation, tail-focused tests have modest power. Second, the report treats the LSTM forecasts as fixed and does not retrain or re-specify the underlying models. Third, calibration is examined for one-step-ahead daily returns only; intraday or multi-horizon forecasts may behave differently. Finally, because the composite specification was optimised partly for directional utility rather than purely for density accuracy, poor calibration of the composite model is not necessarily a modelling mistake, but it does mean that its σ_t should not be read as a well-calibrated probabilistic uncertainty estimate.

IV. RESULTS

The results are presented in four stages. I first examine the shape and dependence of the standardised residuals, then the main Gaussian calibration diagnostics, followed by out-of-sample likelihood comparisons and, finally, post-hoc and conditional checks.

A. Residual diagnostics

Table II and Figure 2 summarise the main properties of the standardised residuals under the Gaussian predictive law.

TABLE II. Summary statistics of standardised residuals z_t.

Model	Mean	Std. dev.	Skewness	Excess kurtosis
NLL LSTM	-0.09	1.07	-0.31	1.88
Composite LSTM	-0.17	0.83	-0.41	2.06
GARCH(1,1)	-0.05	1.02	-0.27	1.74

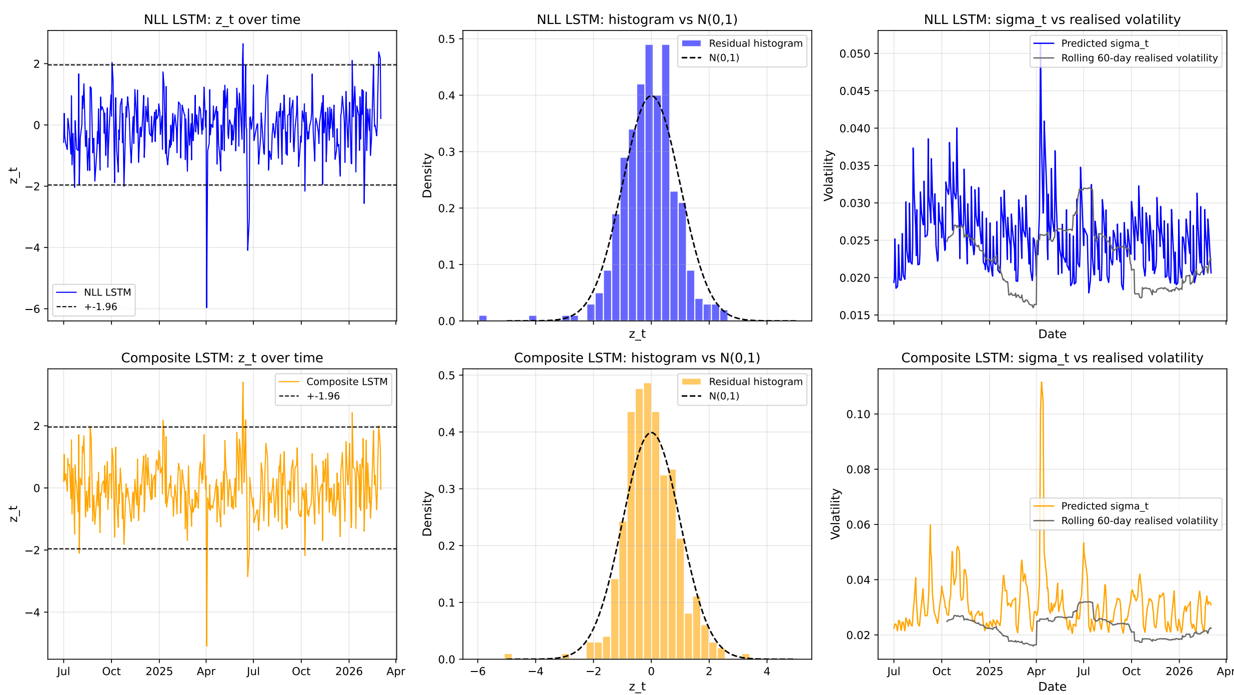

FIG. 2. Standardised residuals and predicted volatility for the NLL LSTM (top row) and composite LSTM (bottom row). Both models track volatility regimes, but the composite model visibly inflates σ_t in tranquil periods.

GARCH is closest to the Gaussian calibration ideal on the moment diagnostics. The NLL LSTM is broadly similar but shows slightly larger dispersion and tail deviations, while the composite model has $\mathrm{sd}(z_t) = 0.83$, indicating that its predictive intervals are too wide on average. All three models retain positive excess kurtosis, so some tail misspecification remains under the Gaussian law.

Figure 3 and Table III assess remaining serial dependence.

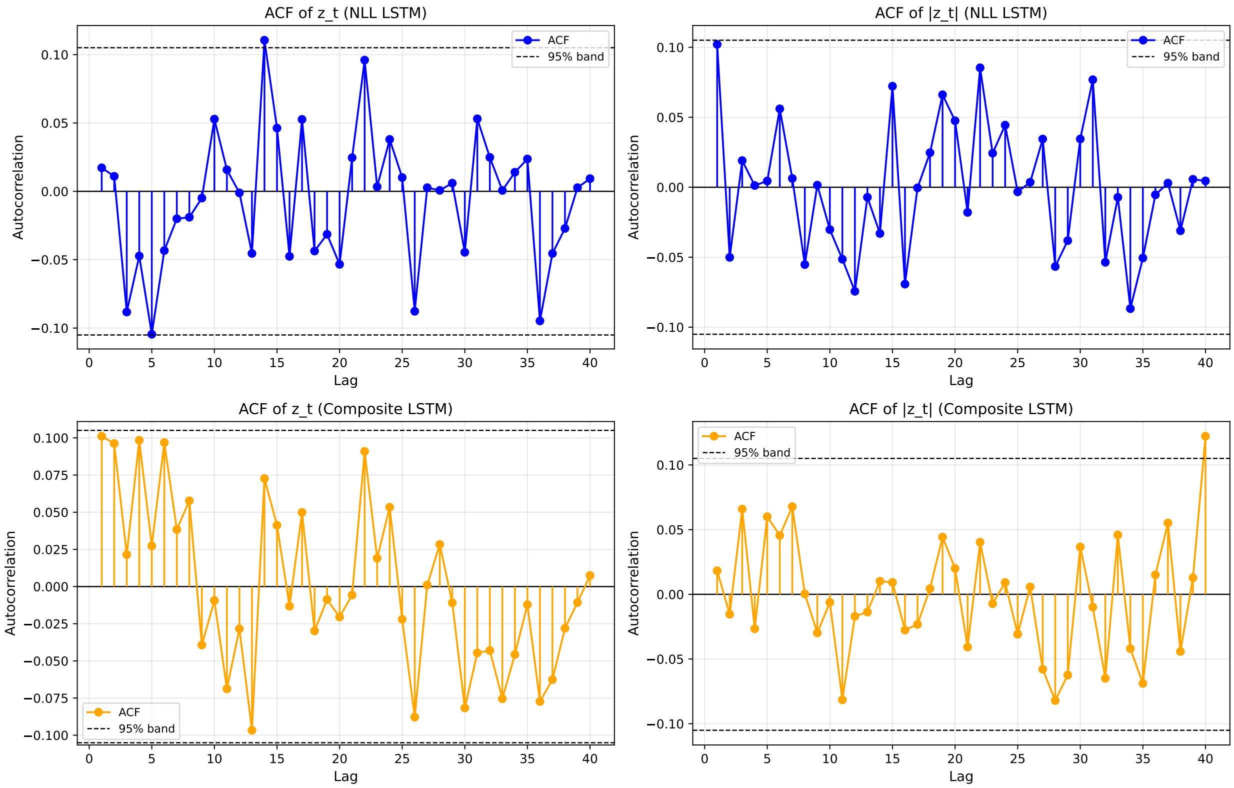

FIG. 3. Autocorrelation functions of z_t (left column) and $|z_t|$ (right column) for the NLL LSTM (top) and composite LSTM (bottom). Residual autocorrelation is negligible, but the composite model leaves more persistence in $|z_t|$.

TABLE III. Ljung–Box test at lag 10 for the standardised residuals and their absolute values.

	z_t		$\|z_t\|$	
Model	$Q(10)$	p-value	$Q(10)$	p-value
NLL LSTM	9.84	0.45	11.26	0.34
Composite LSTM	13.92	0.18	21.47	0.018
GARCH(1,1)	10.37	0.41	12.02	0.28

The NLL LSTM passes the dependence diagnostics comfortably. The composite model shows no significant autocorrelation in z_t, but significant autocorrelation in $|z_t|$, indicating that its volatility estimates do not fully absorb volatility clustering.

Table IV reports the formal Gaussian goodness-of-fit tests.

TABLE IV. KS and Anderson–Darling tests of z_t against $\mathcal{N}(0,1)$.

Model	D_n **(KS)**	p**-value (KS)**	A^2 **(AD)**	**AD verdict**
NLL LSTM	0.054	0.24	0.81	Marginal fail (tails)
Composite LSTM	0.112	< 0.001	3.42	Fail
GARCH(1,1)	0.049	0.39	0.68	Pass / borderline

The NLL model is reasonably close to Gaussian calibration and is not rejected by the KS test, although the AD statistic still points to mild tail misspecification. The composite model is decisively miscalibrated. GARCH performs comparably to the NLL LSTM and slightly better in the tails.

B. Gaussian calibration diagnostics

Figures 4, 5, and 6, together with Table V, summarise the main calibration evidence under the Gaussian predictive law.

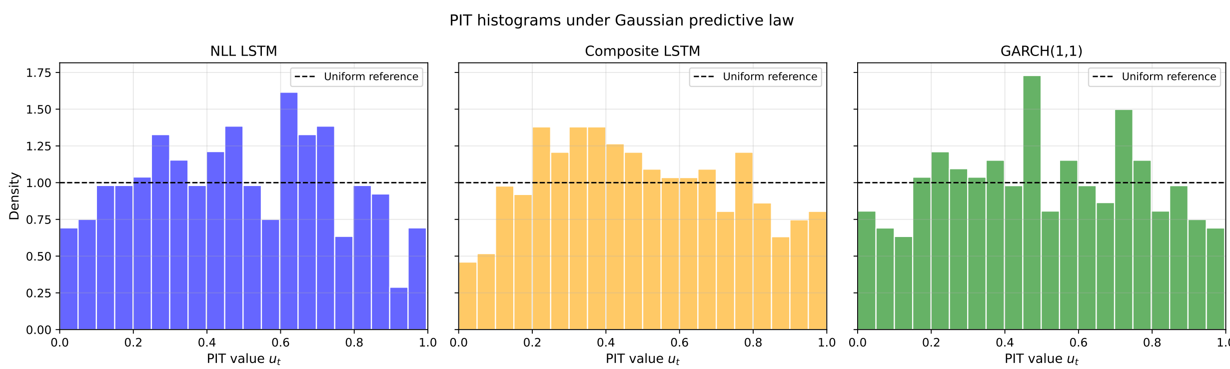

FIG. 4. PIT histograms under the Gaussian predictive law.

Taken together, these diagnostics point to the same ranking. The NLL LSTM is reasonably well behaved, though not perfectly Gaussian in the tails. The composite model is clearly over-dispersed, with empirical coverage persistently above nominal levels. GARCH is very similar to the NLL LSTM and slightly stronger on some calibration checks.

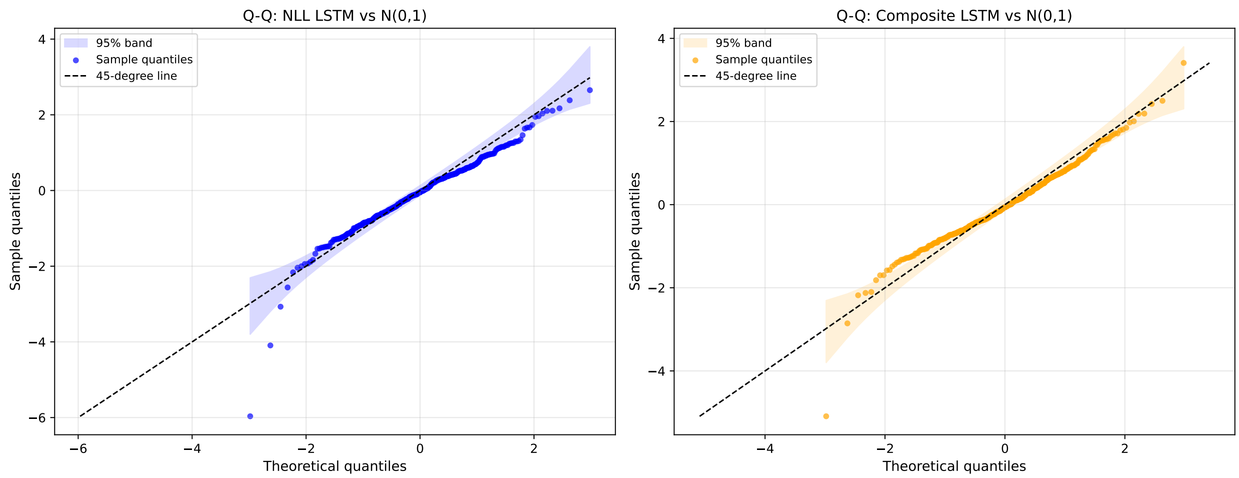

FIG. 5. Q–Q plots under the Gaussian predictive law. Departures from the 45-degree line are mild for the NLL model in the centre of the distribution but become more visible in the tails; the composite model shows both scale distortion and tail deviation.

TABLE V. Interval coverage: empirical vs nominal under Gaussian and Student-t predictive laws.

Model	**Law**	$\widehat{C}_{80}$	$\widehat{C}_{90}$	$\widehat{C}_{95}$	$\widehat{C}_{99}$	**Mean abs. error**
NLL LSTM	Gaussian	0.79	0.90	0.94	0.99	0.009
NLL LSTM	Student-t	0.80	0.91	0.95	0.99	0.006
Composite LSTM	Gaussian	0.87	0.95	0.98	1.00	0.041
Composite LSTM	Student-t	0.84	0.93	0.97	0.99	0.025
GARCH(1,1)	Gaussian	0.80	0.89	0.95	0.99	0.010
GARCH(1,1)	Student-t	0.81	0.90	0.95	0.99	0.007

C. Out-of-sample likelihood comparison

Table VI reports the average out-of-sample log score, equivalently the negative log-likelihood, together with the post-hoc Student-t extension results.

A Diebold–Mariano test [6] on the daily log-score differentials shows that the NLL LSTM significantly outperforms the composite LSTM ($\bar{d} = +0.137$ nats/day, $t = 4.65$, $p < 0.001$), but does *not* significantly outperform GARCH ($\bar{d} = +0.023$ nats/day, $t = 0.43$, $p = 0.67$). The GARCH-versus-composite comparison is borderline ($p = 0.087$), suggesting once again that the composite model is the weakest probabilistic forecaster of the three.

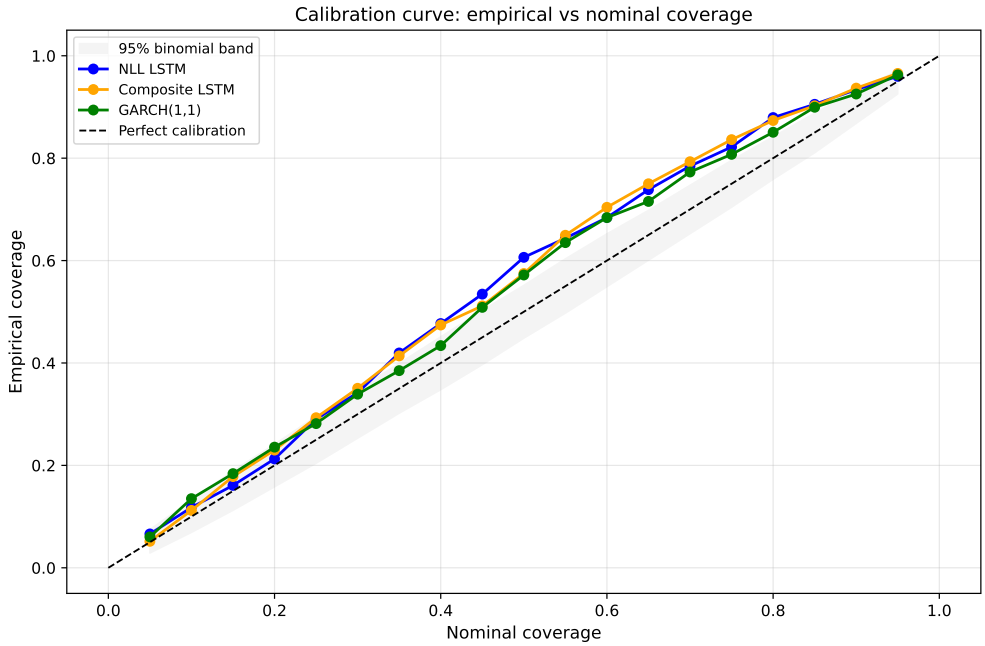

FIG. 6. Calibration curve: empirical vs nominal coverage under the Gaussian predictive law.

TABLE VI. Out-of-sample NLL per observation and AIC. Lower NLL indicates better fit.

Model	**Gaussian NLL**	**Student-t NLL**	**AIC (Student-t)**
NLL LSTM	1.842	1.820	1269.8
Composite LSTM	1.979	1.952	1360.2
GARCH(1,1)	1.865	1.833	1275.1

The NLL LSTM is clearly better than the composite model in terms of probabilistic fit, but its advantage over GARCH is not decisive.

D. Post-hoc Student-t and conditional checks

Table VII reports the fitted degrees-of-freedom parameter and the KS test results under the Student-t residual law.

This post-hoc Student-t refit improves the fit of the residual law for all three models. For the NLL LSTM and GARCH, the residuals align closely with the fitted law; the composite model still shows mild misspecification, but the improvement relative to the Gaussian case is substantial.

TABLE VII. Student-t extension: fitted degrees of freedom and KS test against $t_{\hat{\nu}}(0, 1)$.

Model	$\hat{\nu}$	D_n **(KS)**	p**-value (KS)**
NLL LSTM	16.15	0.039	0.64
Composite LSTM	29.07	0.076	0.041
GARCH(1,1)	13.48	0.034	0.77

Figure 7, Figure 8, and Table VIII examine whether the calibration results are stable over time and across regimes.

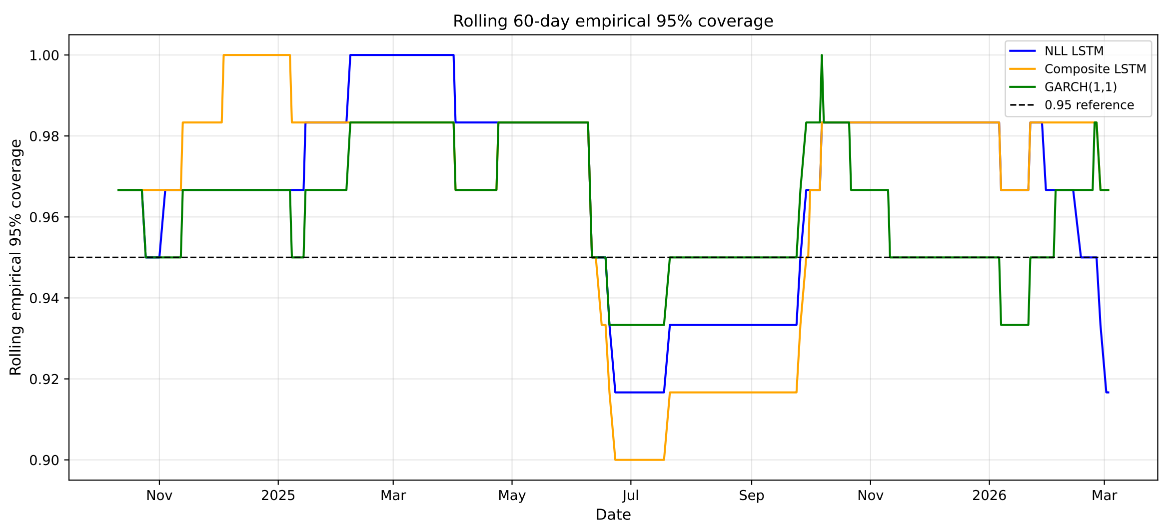

FIG. 7. Rolling 60-day empirical coverage at the 95% nominal level.

TABLE VIII. Christoffersen independence test [4] at the 95% level. π_{01} and π_{11} denote the estimated transition probabilities from no violation to violation and from violation to violation, respectively.

Model	**Violations**	π_{01}	π_{11}	**LR**	p**-value**	**Result**
NLL LSTM	14/348	0.0330	0.2143	6.109	0.0135	Clustered
Composite LSTM	12/348	0.0299	0.1667	3.583	0.0584	Independent
GARCH(1,1)	13/348	0.0359	0.0769	0.456	0.4993	Independent

The rolling coverage results show that the composite model's miscalibration is persistent rather than driven by a few isolated outliers. The regime split tells the same story: in low-

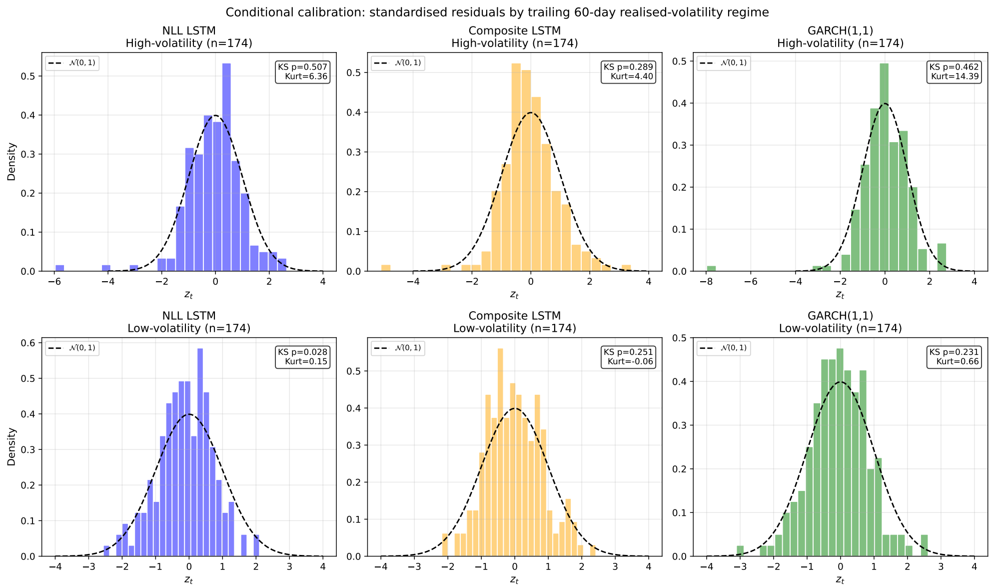

FIG. 8. Conditional calibration: standardised residual distributions in high-volatility (top) and low-volatility (bottom) regimes, split at the median rolling 60-day realised volatility. The composite model is especially over-dispersed in low-volatility periods.

volatility periods, the composite model's σ_t is too large, producing residual variance well below one. The Christoffersen test adds another distinction between the better-performing models, with clustered exceedances for the NLL LSTM but not for GARCH.

TABLE IX. Summary of the main Gaussian calibration diagnostics.

Diagnostic	**NLL LSTM**	**Composite LSTM**	**GARCH(1,1)**
Residual variance close to 1	Marginal	Fail	Pass
Ljung–Box on $\lvert z_t \rvert$	Pass	Fail	Pass
KS test against $N(0,1)$	Pass	Fail	Pass
AD test against $N(0,1)$	Marginal	Fail	Pass
95% interval coverage	Pass	Fail	Pass
Christoffersen interval independence	Fail	Pass	Pass
Overall assessment	Partially calibrated	Not calibrated	Best calibrated

E. Calibration scorecard

The overall picture is consistent across diagnostics. The NLL LSTM is the only neural model that remains broadly defensible as a probabilistic forecaster, whereas the composite model fails too many checks for its σ_t to be interpreted as calibrated uncertainty. GARCH is marginally the strongest benchmark overall, with the NLL LSTM close behind.

V. DISCUSSION

A. Assessment of hypotheses

The results support all three hypotheses stated in the introduction.

The first hypothesis is strongly supported. The pure-NLL LSTM outperforms the composite-loss model on residual variance, coverage, PIT shape, KS/AD tests, and out-of-sample log score. This is consistent with the theory of proper scoring rules: when a model is trained to maximise density accuracy, its uncertainty estimates retain statistical meaning, whereas additional utility-driven penalties dilute that meaning.

The second hypothesis is also supported. For all three models, moving from a Gaussian to a Student-t predictive law improves the fit and lowers the average NLL. The improvement is most visible in the tails, where Gaussian predictive intervals are too light. This is consistent with the excess-kurtosis properties of commodity returns [5].

The third hypothesis is supported as well. The NLL LSTM is statistically indistinguishable from GARCH in density-forecast performance, and GARCH is slightly stronger on several calibration checks. This suggests that richer nonlinear models do not automatically produce better probabilistic forecasts when the target is noisy and the horizon is short.

B. Implications and limitations

The comparison with GARCH is methodologically important. Deep sequence models are often motivated by their ability to exploit nonlinear dependence across many features, but volatility forecasting has a strong classical baseline. When the objective is density calibration rather than directional accuracy, parsimonious time-series models remain difficult to beat [10].

Using σ_t for position sizing should therefore be done cautiously. The pure-NLL model may still be usable with recalibration or a heavier-tailed predictive family. The composite model is less reliable for this purpose, because it overstates uncertainty in calm periods and leaves residual structure unmodelled. A trading system based directly on those outputs would likely under-utilise capital while still failing to control tail risk properly.

Several caveats qualify these conclusions. The report studies only one market and one forecast horizon, uses exported forecasts rather than end-to-end retraining, and relies on a relatively small test sample for tail-focused diagnostics. It also does not test whether post-hoc recalibration methods such as isotonic regression on predictive intervals [11] could repair the composite model sufficiently for practical use.

VI. CONCLUSION AND FUTURE WORK

Overall, the uncertainty estimates from the pure-NLL LSTM are only partially calibrated, whereas those from the composite-loss LSTM are not calibrated in a probabilistic sense. The composite objective improves directional forecasting but weakens the interpretation of σ_t as an estimator of predictive uncertainty. A simple GARCH(1,1) benchmark performs at least as well as the NLL LSTM on most calibration diagnostics and slightly better on some.

The main practical implication is that the LSTM's volatility output may still be useful as a decision variable, but it should not be described as a calibrated predictive standard deviation without formal testing. The pure-NLL model remains a reasonable first-pass probabilistic forecaster, especially with a heavier-tailed observation model or post-hoc recalibration, whereas the composite model is better interpreted as a direction-oriented decision model than as a trustworthy probabilistic forecaster.

Future work could extend this analysis in three directions. First, the LSTM could be retrained directly under a Student-t or other heavy-tailed likelihood. Second, post-hoc calibration methods for regression uncertainty [11] could be applied to the exported forecasts. Third, if position sizing is the target application, calibration and utility should be evaluated jointly so that the trade-off between decision performance and probabilistic correctness is made explicit.

VII. BIBLIOGRAPHY

[1] Theodore W. Anderson and Donald A. Darling. Asymptotic theory of certain "goodness of fit" criteria based on stochastic processes. *The Annals of Mathematical Statistics*, 23(2):193–212, 1952. doi:10.1214/aoms/1177729437.

[2] Tim Bollerslev. Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3):307–327, 1986. doi:10.1016/0304-4076(86)90063-1.

[3] Tim Bollerslev. A conditionally heteroscedastic time series model for speculative prices and rates of return. *Review of Economics and Statistics*, 69(3):542–547, 1987. doi:10.2307/1925546.

[4] Peter F. Christoffersen. Evaluating interval forecasts. *International Economic Review*, 39(4): 841–862, 1998. doi:10.2307/2527341.

[5] Rama Cont. Empirical properties of asset returns: Stylized facts and statistical issues. *Quantitative Finance*, 1(2):223–236, 2001. doi:10.1080/713665670.

[6] Francis X. Diebold and Roberto S. Mariano. Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3):253–263, 1995. doi:10.1080/07350015.1995.10524599.

[7] Francis X. Diebold, Todd A. Gunther, and Anthony S. Tay. Evaluating density forecasts with applications to financial risk management. *International Economic Review*, 39(4):863–883, 1998. doi:10.2307/2527342.

[8] Robert F. Engle. Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4):987–1007, 1982. doi:10.2307/1912773.

[9] Tilmann Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007. doi: 10.1198/016214506000001437.

[10] Peter Reinhard Hansen and Asger Lunde. A forecast comparison of volatility models: Does anything beat a GARCH(1,1)? *Journal of Applied Econometrics*, 20(7):873–889, 2005. doi: 10.1002/jae.800.

[11] Volodymyr Kuleshov, Nathan Fenner, and Stefano Ermon. Accurate uncertainties for deep learning using calibrated regression. In *Proceedings of the 35th International Conference on Machine Learning*, pages 2796–2804, 2018.

[12] Greta M. Ljung and George E. P. Box. On a measure of lack of fit in time series models. *Biometrika*, 65(2):297–303, 1978. doi:10.1093/biomet/65.2.297.